

Handwriting Recognition in VR for Enhanced Learning and Immersive Interaction Experience

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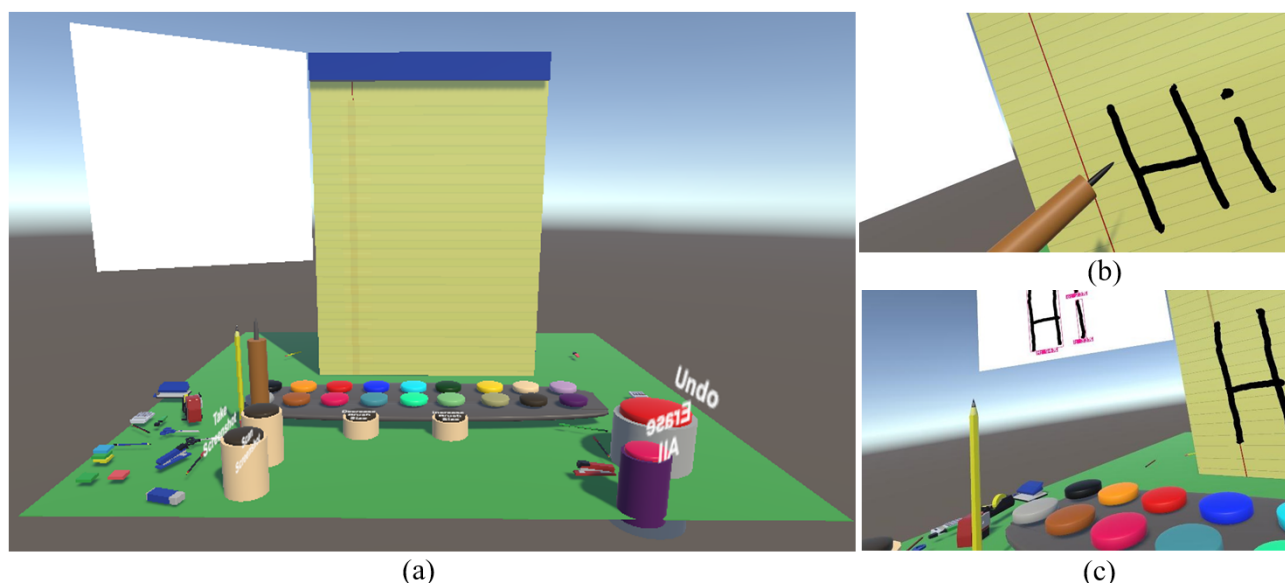


Fig. 1: (a) The virtual workspace where users can write on a notepad with a pen in different colors and line widths. (b) The user writes “Hi.” (c) The basic optical character recognition (OCR) model parses each character and displays the results on a whiteboard next to the notepad, with confidence scores shown in pink at the bottom of each character. The confidence score for “H” is 94%. The confidence score for the lowercase letter “i” was generated in two parts: the bar of the lowercase “i” received a confidence score of 33%. The dot in “i” was misclassified as the lowercase “o” with a confidence score of 28%.

Abstract—In this research, artificial intelligence (AI) and virtual reality (VR) are combined to address challenges in handwriting. A pipeline has been developed to enable users to practice handwriting within their individualized VR workspace. Next, the texts were parsed by a handwritten text recognition (HTR) model. The proposed approach has demonstrated the potential to motivate learners to practice handwriting in VR and could also serve as a valuable tool for mastering a foreign language. With a robust HTR model, this approach also facilitates tracking learning progress through AI-generated feedback. In addition to enhancing learning, the proposed approach can serve as an early intervention and individualized management strategy for individuals experiencing dysgraphia-related challenges.

Index Terms—handwriting recognition, artificial intelligence applications, virtual reality

I. INTRODUCTION

The virtual environment (VE) is more dynamic, interactive, and scalable compared to the real world [1]. VR learners felt

3.75 times more emotionally connected to the content than classroom learners [2]. The interactive and immersive nature of VR can spark curiosity, fun, and intrinsic motivation among learners, leading to a deeper exploration of content [3]–[5]. Hence, learning can be reinforced with VR.

Handwriting has a significant impact on cognition, memory development, and memory retention, shaping the overall learning experience from childhood [6]. Although technology is prevalent, handwriting offers unique neurological benefits relative to typing. Writing by hand activates multiple brain regions involved in language learning [7]. This brain activity is higher during handwriting than during typing, thereby strengthening neural connections. For young children, this process is crucial to the brain’s functional development. Hence, handwriting is not merely an antiquated skill but a vital cognitive tool that supports learning, brain development, and practical communication skills throughout life.

The best way to learn a language is to immerse oneself in it, which includes mastering it through diligent handwriting practice. An application for learning a language on an immersive platform, such as VR, may be more appealing than simply writing vocabulary in a notebook. With recent advances in handwritten text recognition (HTR) models, a virtual tutor or avatar can guide learners on their language-learning journey by providing feedback that supports self-evaluation and motivates practice.

Handwriting recognition in VR by deploying different HTR models is not new. With robust HTR models, handwriting in VR can be successfully transferred to facilitate different VR activities [8]. Various ways to write in VR, using different interfaces, were also explored to understand user preferences [9]. VR enables users to modify an environment to meet their requirements in ways not possible in the real world.

However, handwriting in VR has yet to be explored for language learning. VR can also be used to create real-world scenarios in which learners can test their skills with an AI tutor powered by a robust handwriting recognition algorithm, without help from others. This research uses VR as a personalized platform for learners to practice handwriting. A robust handwriting recognition algorithm allows users to monitor progress as they continue their language learning journey.

Our proposed hands-on learning approach will create an affective learning environment, so that learners can connect, understand, and face challenges in a more individualized way than the traditional approach. Learners will explore various interfaces for writing in VR. We will foster a sense of belonging among learners by enabling them to develop their personalized learning platforms. User studies will be conducted to test our hypothesis and understand how practicing handwriting in VR, along with a robust AI tutor that monitors learners' progress, affects the language-learning experience. In addition to learning enhancements, this research could benefit people with dysgraphia, a neurological disorder that affects writing. Currently, there are no medications specifically for its treatment. However, early intervention and individualized management strategies can help mitigate challenges associated with dysgraphia. Different interfaces, along with a personalized learning space for practicing handwriting, may help people with similar conditions improve memory and overcome challenges associated with pen-based writing.

II. BACKGROUND

VR facilitates feelings of immersion and presence and has been reported to improve task-specific performance across various contexts [10], [11]. Hence, handwriting in VR can be more effective than in the real world, as learners are more motivated to continue writing for longer. A robust handwriting algorithm can inform learners about their performance, as improved handwriting increases the likelihood of automatic recognition.

Different training models for handwriting recognition were developed by scanning images of written documents and demonstrated proficiency across various contexts. Tesseract

OCR [12], [13], an Optical Character Recognition (OCR)-based approach, uses Long Short-Term Memory (LSTM) models. LSTMs excel at remembering long-term dependencies, which can be used for prediction, and are more accurate in classifying letters written in sequence. EasyOCR uses a CRNN [14] architecture. A CRNN is a combination of a Convolutional Neural Network (CNN) and a Recurrent Neural Network (RNN). The CNN component allows the model to identify individual letters/texts, and the RNN helps recognize the sequence throughout the document based on previous occurrences. TrOCR [15] is another OCR-based handwritten text recognition model (also known as the transformer model) that analyzes the image/document all at once and weighs different parts with levels of importance. This allows the model to piece together various parts of the document with higher accuracy; i.e., the model is more likely to identify the dot above an "i" as part of the letter rather than a stray period.

Researchers explored various VR interaction interfaces to identify user preferences. Collision-based interactions have received greater appreciation than raycast-based approaches [16]. With a haptic device, users feel force feedback from the virtual board as they continue writing on it. The option to choose from several interfaces provides learners with greater flexibility to practice handwriting in their preferred mode.

This research aims to leverage the immersive power of VR to encourage learners to practice handwriting in their personalized virtual space. A robust HTR model helps them track their progress by providing real-time feedback on the accuracy of handwritten text. The proposed approach will allow users to practice handwriting in their personalized VR space, using colorful inks, erasers, and notepads. Additionally, users will explore different interfaces, e.g., writing on a notepad using a VR controller, either by colliding with a pen or casting a ray. We will explore VR's hand-tracking capabilities to enable learners to write in VR with their hands. The handwritten texts are saved as images and next, fed to various HTR models. A robust HTR model provides feedback of confidence levels, which can be reclassified as correct, nearly correct, partially correct, and incorrect. Based on the feedback, learners would continue practicing.

Handwriting recognition is an evolving field; robust HTRs are being developed. Some are trained for character recognition, while others are made for word and sentence recognition. Some recent HTR models, such as TrOCR, reportedly achieved a very high accuracy in handwriting detection. Based on feedback from various training models and their accuracy in handwritten character, word, and/or sentence recognition, we aim to select the model(s) that perform well for virtual tutors. A personalized platform that incorporates multiple interactive techniques would foster an inclusive learning experience for learners across the spectrum.

The next section describes the proposed approach, followed by the integration of a popular HTR model and initial findings from the pilot study. Next, our concluding remarks and scope for future work followed. The pilot study demonstrates VR as a favorable platform for handwriting practice and suggests

that it could be developed into a full-fledged VR-enhanced language-learning tool, with scope for self-evaluation and learner motivation.

III. THE PROPOSED APPROACH

Based on the flowchart shown in Figure 2, the proposed approach can be divided into three main parts:

- Handwriting in VR,
- Scanning of the handwritten texts by HTR models, and
- Evaluating learning progress based on the output from the HTR model

A. Handwriting in VR

In a VR-enhanced virtual environment (VE), learners write on a notepad using their preferred interface. As shown in Figure 2a, the proposed approach uses both VR controllers and the haptic device interface for handwriting in VR. Both collision-based and non-collision ray casting techniques will be used. With collision-based writing, the pen tip collides with the notepad. The VR controller does not provide force feedback to users when the pen tip collides with the notepad. However, with a haptic device, users experience force feedback from the notepad as they press the pen tip against it to start writing, mimicking the real-life experience. In this case, the virtual pen follows the haptic device handle's movement.

With the non-collision ray-casting technique, users write by casting a ray from the VR controller onto the notepad. With the hand-tracking approach, users' fingers will be tracked for VR interaction, allowing them to write more freely, simply coordinating hand and finger movements without grabbing the VR controller. Figure 2a shows different interfaces to be adopted by users for handwriting in VR.

Users write characters, words, and sentences in VR. They can increase or decrease line width while writing. They can erase the writing and start afresh on a clean notepad. They can write by interactively choosing colors from a color palette.

B. Scanning of the handwritten texts by HTR models

In VR, users write characters, words, and sentences using different interfaces and techniques, as discussed above. Handwritten text on a virtual page is saved as an image, which is then scanned by an HTR model. Figure 2 b shows this step. As discussed in the background section, different HTR models demonstrate proficiency across different categories: some achieve higher accuracy with characters, while others perform better with words. Different HTR models will be continuously integrated and tested, and the one that produces the optimal output will be used.

C. Evaluating learning progress based on the output from the HTR model

This step comprises two parts: a confidence score generated by an HTR model, followed by an evaluation of the learner's performance. The overall HTR model confidence score is divided into four categories, each mapped to different performance levels to facilitate self-evaluation. Figure 2 c illustrates this step, showing the confidence score categories and the corresponding user performance levels.

IV. IMPLEMENTATION

The project was implemented in Unity3D. Unity's XR plug-in facilitates VR integration. Unity's free asset, "OpenCV plus Unity" (<https://assetstore.unity.com/packages/tools/integration/opencv-plus-unity-85928>) includes a plug-in

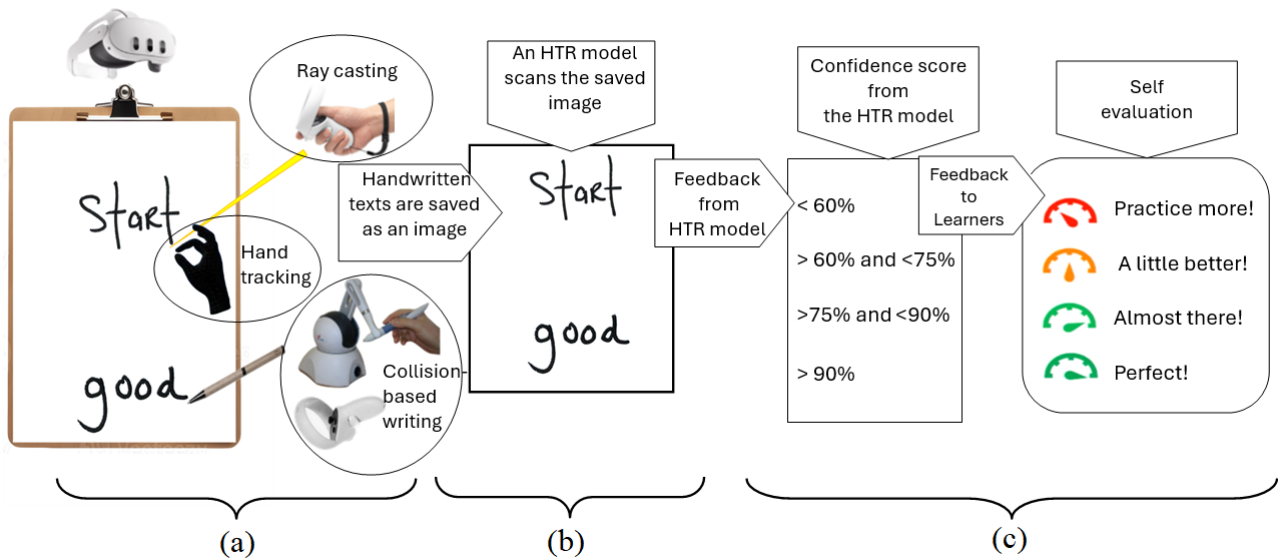


Fig. 2: (a) Different interfaces to write on a virtual notepad in VR. (b) The handwritten text is saved as an image for scanning by an HTR model. (c) The HTR model generates confidence scores, which are mapped to user performance levels for self-evaluation.

for the OCR model that works with both typed and handwritten text. Our pilot study uses this HTR model. After the user writes in VR, the handwritten text is parsed by the OCR model.

Figure 1a shows the VR setup for handwriting. Users can write on the yellow notepad with a virtual pen when the pen tip collides with the notepad. In our pilot study, we used a collision-based approach with the VR controller. The OCR model's output will be displayed on the whiteboard next to the notepad. Unity's line renderer draws line strokes as users continue writing. Those line strokes are saved in a stack. Users can choose an ink color from the palette by dipping the pen tip into it.

There are several virtual buttons in the scene. Figure 3 shows different buttons with labels. When the virtual hand collides with a button, the button is pressed down to perform the corresponding action.

As users write on the notepad, they can increase or decrease the line width by pressing the "Increase Line Width" and "Decrease Line Width" buttons, respectively. The "Erase All" button allows users to erase all handwriting from the virtual notepad to start afresh. All line strokes stored in the stack are removed, and the stack becomes empty. The "Undo" button allows users to correct their writing by partially erasing content. Partial erasing works by popping line strokes from the stack one by one, starting with the very last one. After completing the writing, users press the "Take Screenshot" button to save the handwritten text as an image. This has been accomplished by introducing a second camera, the "Screenshot" camera, into the scene. This second camera filters out other objects in the scene except lines drawn by the line renderer. Next, users press the "Scan Screenshot" button. The integrated OCR model scans the handwritten text, and the result is displayed on the whiteboard next to the yellow notepad. The results of the OCR integration with handwritten text in VR are described and analyzed in the next section.

V. EVALUATION

A. User Study

The project is currently in its work-in-progress phase. It was implemented as part of a master's thesis. After developing a VR handwriting pipeline that supports simultaneous scanning with OCR, the authors conducted a feasibility study of the prototype by writing a few characters and words. An Oculus Quest 3 headset was used for VR writing.

B. User Experience in VR

Handwriting in VR was more dynamic, playful, and less constrained than real-life pen-and-paper handwriting. However, VR did not provide the real-life force feedback and surface friction we experience while writing on paper with a pen. We were more cautious while handwriting in VR, and our writing speed was slower than in real life. We experienced limited controller accuracy compared to the pen. Increased line width produced better stroke quality, as fewer artifacts were observed in line strokes.

C. Results and Analysis

Figure 4 and Figure 5 show the OCR-generated results for the handwritten texts in VR, with confidence scores in pink beneath each character. As shown in Figure 4, among the four iterations, "A" was correctly recognized once, with a confidence score of 62%. Two times "A" was mistaken for "X" with confidence scores of 41% and 39%, respectively. "A" was misclassified as "V" one time and the confidence score was 46%. The letter "O" was correctly detected in all iterations. However, the confidence scores were low, ranging from 29% to 33%. "S" was detected correctly three times out of four iterations, and the confidence scores ranged between 45% and 48%. In the other case, S was recognized as "m" with a confidence score of 69%. 'D' was mistaken for "O" in all four iterations, and the confidence scores ranged from 13% to 26%. In the word "Hi," the letter "H" was recognized correctly twice, with confidence scores of 53% and 94%. The other time it was recognized as the lowercase "h," with a confidence

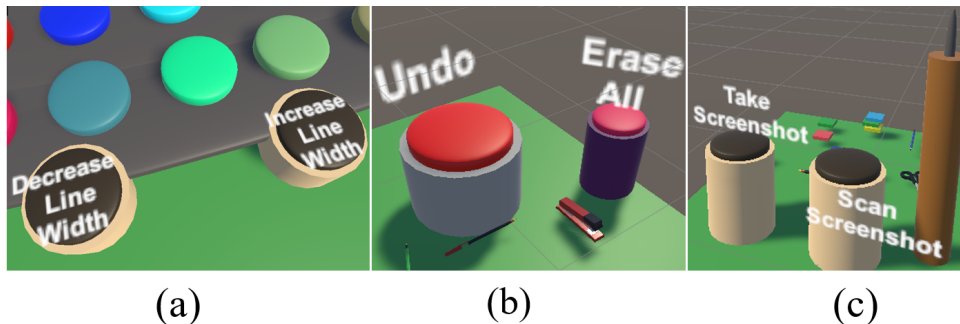


Fig. 3: (a) Buttons for increasing and decreasing line widths ("Increase Line Width" and "Decrease Line Width" buttons, respectively). (b) Buttons for partially and completely erasing handwriting ("Undo" button and "Erase All" button, respectively). (c) Buttons for saving the handwritten text as an image and scanning it by the OCR model "Take Screenshot" and "Scan screenshot", respectively).

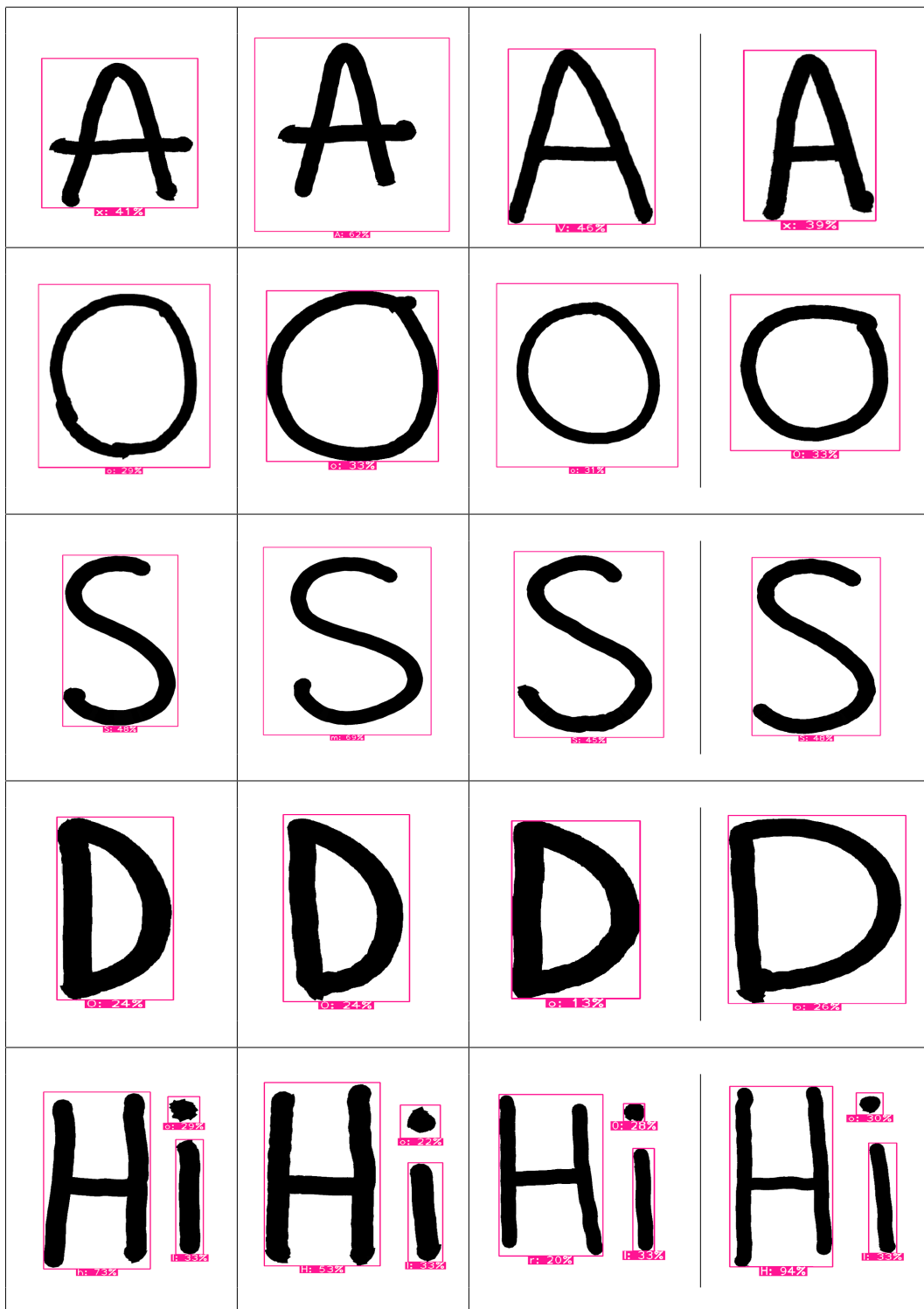


Fig. 4: The results generated by the OCR model for characters/words handwritten in VR with corresponding confidence scores in pink.

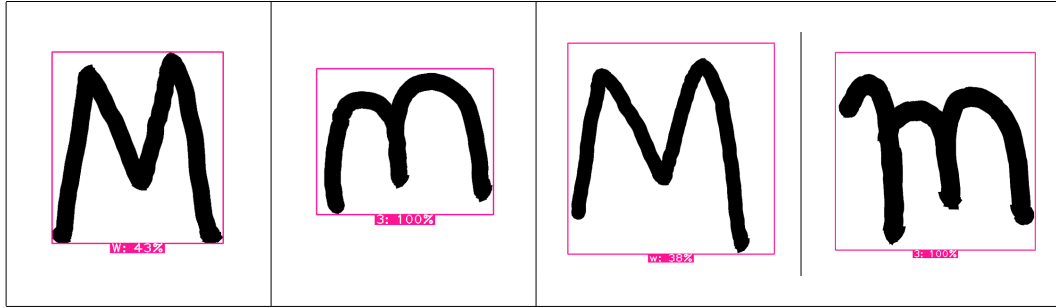


Fig. 5: The results generated by the OCR model for “M” handwritten in VR with corresponding confidence scores in pink.

score of 73%. One instance of “H” in “Hi” was misclassified as “r” with a confidence score of 20%. The letter “i” was recognized in two parts. In all cases, the bar in the lowercase “i” was correctly recognized, yielding a confidence score of 33% in all cases. However, the dot was classified separately and misclassified as “o”, with confidence scores ranging from 22% to 30%. The letter “M” was identified as the digit “3” two times with a 100% confidence score. This happened because OCR encloses characters and numbers in a tight-fitting box and then compares them to the nearest match. Because OCR considers character/digit orientation during comparison, the lowercase “m” was matched to the digit “3” rotated anticlockwise by 90 degrees. The other two times “M” was misclassified as “W” with confidence scores of 38% and 43%. In those cases, OCR found a closer match for “M” (inverted) with “W” than other characters.

The OCR model correctly identified some characters; however, it showed low confidence for most of them. Some characters, like “A,” “D,” and “M” were identified incorrectly. Given the confidence scores in our proposed design and the one-to-one mapping to learners’ progress, OCR would provide very misleading information.

D. VR Handwriting with Other HTR Models

Although OCR did not produce the optimistic output required for our approach to learners’ self-evaluation, it helped us build the initial platform. At the time of writing, we wrote twenty words in VR using our approach and tested them separately with the TrOCR model [15], a recent HTR model described in the background section. Figure 6 shows the dataset generated from VR writing using our approach. ChatGPT generated the 20-word dataset with the following prompt to evaluate different HTR models: “Give me 20 random words to handwrite in VR.” Given the stroke complexity and the mixture of uppercase and lowercase letters, the dataset was appropriate for assessing the robustness of HTR models and user resilience in VR handwriting. Each word also has sufficient width to challenge learners’ ability to maintain a straight baseline without a physical sheet of paper.

We received remarkable results with the TrOCR model. Figure 7 presents the results. TrOCR correctly identified 19 words out of the 20 words from our dataset. We are integrating TrOCR into our pipeline to replace the current OCR model.

VI. CONCLUSION AND FUTURE WORK

The proposed approach enables users to practice handwriting in VR, a platform that can motivate learners to continue

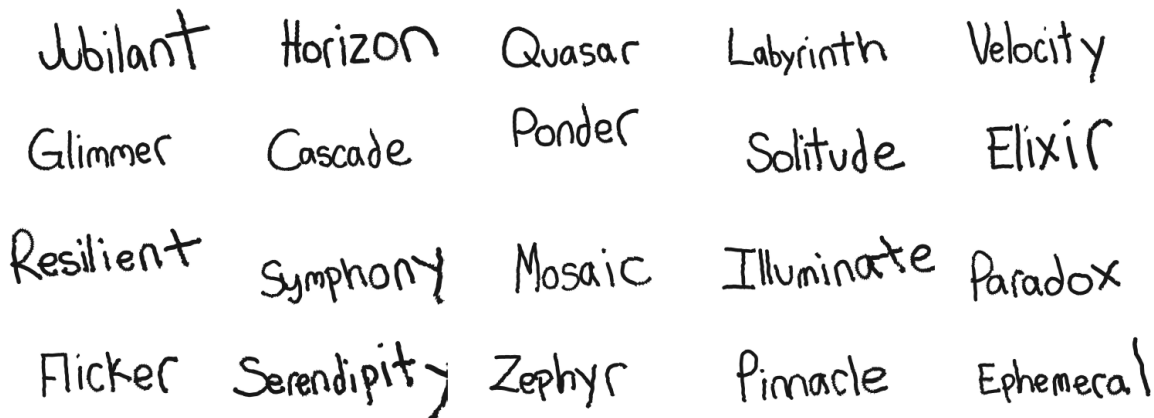


Fig. 6: The 20-word dataset handwritten in VR with our approach.

TrOCR Model with VR Handwriting images	
File: Alphabet1.png, Predicted Text: '['subitant']'	• Jubilant
File: Alphabet10.png, Predicted Text: '['2 Solitude']'	• Solitude
File: Alphabet11.png, Predicted Text: '['2 Elixir .']'	• Elixir
File: Alphabet12.png, Predicted Text: '['Resilient']'	• Resilient
File: Alphabet13.png, Predicted Text: '['2 Symphony']'	• Symphony
File: Alphabet14.png, Predicted Text: '['1 Mosaic .']'	• Mosaic
File: Alphabet15.png, Predicted Text: '['2 Illuminate .']'	• Illuminate
File: Alphabet16.png, Predicted Text: '['paradox']'	• Paradox
File: Alphabet17.png, Predicted Text: '['2 Flicker .']'	• Flicker
File: Alphabet18.png, Predicted Text: '['Serendipity']'	• Serendipity
File: Alphabet19.png, Predicted Text: '[' - Zephyr']'	• Zephyr
File: Alphabet2.png, Predicted Text: '['1 Horizon .']'	• Horizon
File: Alphabet20.png, Predicted Text: '['2 Pinnacle .']'	• Pinnacle
File: Alphabet21.png, Predicted Text: '['2 Ephemeral']'	• Ephemeral
File: Alphabet3.png, Predicted Text: '['2 Quasar .']'	• Quasar
File: Alphabet5.png, Predicted Text: '['1 Labyrinth .']'	• Labyrinth
File: Alphabet6.png, Predicted Text: '['2 Velocity']'	• Velocity
File: Alphabet7.png, Predicted Text: '['6 Glimmer .']'	• Glimmer
File: Alphabet8.png, Predicted Text: '['2 Cascade .']'	• Cascade
File: Alphabet9.png, Predicted Text: '['2 Ponder']'	• Ponder

Fig. 7: TrOCR [15] detected 19 words correctly out of the 20-word dataset handwritten in VR with our approach.

writing for longer than in the real world. As mentioned earlier, the practice of writing in VR using different interfaces would allow learners from diverse backgrounds to use the platform. Instead of line renderers, color blobs may produce better stroke quality and control during VR handwriting. We will test the prototype with color blobs of varying radii. As part of future work, we will gradually integrate different interfaces and techniques as discussed. A haptic interface promises to provide a lifelike experience by generating force feedback during VR writing. The proposed approach would allow people with some level of disability, such as dysgraphia or other neurological disorders, to practice handwriting. They can try different interfaces that would work for them, which can positively affect their lifestyles in the long run. As we continue to integrate various interfaces, we will conduct a usability study with learners with and without disabilities. Future evaluations will focus on user engagement, preference, motivation, and performance.

It can be hypothesized that a robust handwriting algorithm can inform learners about their performance, as improved handwriting increases the likelihood of automatic recognition. Although OCR is not suitable for self-evaluation, models such as TrOCR have demonstrated potential for handwritten text, and we are working to replace the current OCR model with TrOCR.

Handwriting recognition is a rapidly evolving field; newer HTRs with more robust features are being developed. We will continue to explore various HTR models for handwritten text

using our VR platform, aiming to create a comprehensive handwriting application that not only allows users to practice on their individualized platform but also supports self-evaluation, enabling them to track improvements.

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